

Production lines — Detailed design

005_DD implements SCR-005, FN-032–FN-036 and REQ-064–REQ-069 under WI-009 approved plan revision 1, step 6. This is the main DD review package, not the completed DD family or an implementation authorization.

1. Document control

Field	Value
Document ID / version	005_DD / 1
Category / system / subsystem	UI and module design / ProductionManagementAI / Master data
Screen	SCR-005 「生産ライン・工程」 (Production lines and processes)
Work item / author	WI-009 / Agent
Created / updated	2026-10-01 / 2026-10-01
Inputs	Approved 005_BD version 1 and 005_DB version 1; DEC-001–DEC-009
State	Submitted for review; not implemented

Version	Date	Author	Change
1	2026-10-01	Agent	Initial fields, modules, states and persistence boundaries

2. References, companions and ownership

Document	Purpose / current relationship
005_BD	Approved routes, regions, navigation and business behavior
005_DB	Approved schema, exact bounds, unit revision, locks and migration intent
005_REQ	Acceptance criteria for REQ-064–REQ-069
WI-009_plan	Sequential review and design-only authorization
Layering_ADR	Controller/Application/Domain/Infrastructure boundaries

Document	Purpose / current relationship
Authentication ADR	Existing cookie-session and role foundation

Companion design documents

All three companions are mandatory for the completed DD family. They have not yet been authored because the approved plan requires separate review in order. Names below are reservations, not links to nonexistent files. This main document owns fields, screen modules and states; exact catalogs/algorithms have one home.

Reserved document	Owns	Next step
005_DD-API_生産ライン・工程.md	Requests/responses, query bounds, roles, errors and operation telemetry	7, after main DD review
005_DD-FN_生産ライン・工程.md	Application methods, lock SQL/order, atomic writes and backend instrumentation	8, after API review
005_DD-SPD_生産ライン・工程.md	Client processing sequences, detailed SVGs and EN/JA HTML state mockups	9, after FN review
005_DD-ORD_production-line-assignment.md	SCR-001/002 effects and existing-order request compatibility	10, after SPD review

Referenced by: none yet. Later companions will reference this main DD; no edits to approved source designs are required. Overall family consistency stays pending until all companions and the new order-impact DD are reviewed.

Task and module index

No.	Task / function	Design owner	Review status
1	List and URL filters, FN-032	Main DD fields/states; SPD sequences; API read catalog	Main DD submitted; companions pending
2	Identity/hours form, FN-033	Main DD validation; API/FN mutation details	Main DD submitted; companions pending
3	Product timing/reconfirmation, FN-034	Main DD row state; DB constraints; API/FN confirmation guard	Main DD submitted; companions pending
4	Line retirement, FN-035	Main DD dialog/state; API/FN transaction	Main DD submitted; companions pending
5	Order eligibility/locking, FN-036	DB boundary and new order-impact DD	Detailed order behavior pending step 10

3. Component organization and screen-owned modules

X-1. Paths

Planned path	Responsibility
src/frontend/src/features/production-lines/	Pages, timing editor, decimal validation and API adapter
src/frontend/src/features/production-orders/messages.ts	Japanese catalog additions; one existing catalog, no new i18n dependency
src/frontend/src/components/icons.ts	Centralized Factory import, distinct from Package and ClipboardList
src/backend/ProductionManagementAI.Domain/	Line/pair entities and immutable identity invariants
src/backend/ProductionManagementAI.Application/	Use cases and ports; methods owned by DD-FN
src/backend/ProductionManagementAI.Infrastructure/	EF mapping, repositories and planned additive migrations
src/backend/ProductionManagementAI.Api/	ProductionLinesController; contracts owned by DD-API

X-2. Shared components

Reuse AppHeader/AppNavbar, GuardedLink, NavigationGuardProvider, same-origin apiClient, session handling and Japanese display formatting. Native dialog behavior must retain WI-008 viewport centering and focus behavior. Existing Product master version serialization remains unchanged; the new line token is an opaque string. Framework/ASPX include files and XML rule engines are not applicable to this React/.NET stack.

X-3. Feature modules

Proposed modules below are design names, not existing or tested files. All are created/updated by Agent on 2026-10-01. UI modules return React elements; controller returns HTTP results; validators return structured field errors. Plain rendering and controller modules have no rule-engine parameters.

Module	Inputs / preconditions	Responsibility / output	Dependencies
ProductionLinesPage	Authenticated route; URL q/state/page	Applied-filter list state, PC table/SP cards, notices and pagination	Line API adapter, header, catalog
ProductionLineFormPage	Create or edit id; authoritative detail when editing	Immutable baseline, local draft, version, dirty guard and save outcome	TimingEditor, validation, API adapter
TimingEditor	Saved/new rows; product unit/revision; permitted writer	Add, edit coefficient, stage retirement and explicit confirmation intent	Product choices, catalog, decimal validator

Module	Inputs / preconditions	Responsibility / output	Dependencies
LineConfirmationDialog	Named target/action and invoker	Cancel/Escape safe; confirm action; focus restoration	Native dialog, form/list state
LineStateLabel	Active state and optional stale-unit state	Visible Japanese text and non-color cue	Catalog
ProductionLinesController	Authenticated request; role for writes	Delegate validated use case, map errors without SQL/internal details	Application ports in DD-FN; contracts in DD-API

X-4. External APIs

Only existing same-origin application APIs; see section 8. No external identity, shop-floor API or new integration. Service methods and step-by-step processing are reserved to FN/SPD; this document does not duplicate their algorithms.

X-5. Responsive composition

One data/state implementation for PC table and SP cards; shared forms reflow without a separate mobile workflow. Preserve approved BD breakpoints/regions. The later SPD refines visual details without changing this route/state contract.

Screen-owned validation rule

LineDraftValidation is a format and cross-field assistance rule. Arguments: mode, saved baseline, local identity/hours and timing rows. References: BD screen items/history rules and DB sections 3–4. Checks: code/name strings, exact decimal text, unique product IDs and explicit current-unit intent. Output: field errors indexed by stable product ID or unsaved row key, plus summary. Dependencies: approved product unit set and server-supplied current unit/revision. Local validity does not prove current eligibility or authorize a write; server repeats checks. No database reads or automatic unit conversion occur in this client rule.

4. Screen layout and region mapping

The approved BD SVGs are reused unchanged for this main-DD review. The plan reserves detailed wireframes and EN/JA HTML state mockups to step 9 (SPD); they are not claimed complete here. No new layout or numbered region is introduced.

① **ProductionManagementAI**

ダッシュボード 製造指示一覧 製品マスタ 生産ライン・工程

② **生産ライン・工程**

新規ライン

③ **検索**

ラインコード・ライン名 状態 検索 クリア

④ **一覧**

コード ライン名 稼働時間／日 状態

L-001 第1ライン 8時間 使用中

編集 使用停止

⑤ **ページ・状態**

全件数 前へ 1 次へ

⑥ **生産ライン編集**

ラインコード：L-001（変更不可）

ライン名：第1ライン 稼働時間／日：8

保存 キャンセル

⑦ **生産可能な製品**

製品 単位 製造時間／単位（分）

P-1001 個 2.5 使用中

製品を追加 使用停止

⑧ **単位の確認が必要です**

前の単位：個 現在の単位：kg

現在の単位で確認

⑨ **確認ダイアログ**

このラインを使用停止にしますか？

キャンセル 使用停止

① PMAI
メニュー

② 生産ライン・工程
新規ライン

③ 検索
ラインコード・ライン名
状態 検索 クリア

④ 一覧カード
L-001 第1ライン
稼働時間／日：8 時間 使用中
編集 使用停止

⑤ ページ・状態
全件数 前へ 1 次へ

⑥ 生産ライン編集
ラインコード：L-001 (変更不可)
ライン名：第1ライン
稼働時間／日：8 保存 キャンセル

⑦ 生産可能な製品
P-1001 単位：個
製造時間／単位：2.5 分／個
製品を追加 使用停止

⑧ 単位の確認が必要です
前の単位：個 現在：kg
現在の単位で確認

⑨ 確認ダイアログ
使用停止しますか？
キャンセル 使用停止

BD region	DD items / behavior
1	Shared header/nav and current marker; Factory decorative, visible text

BD region	DD items / behavior
2	Mode heading, New and dirty/permission notices
3	q/state draft controls and applied URL filters
4	Code/name/hours/state and row actions; table/cards share data
5	Loading/empty/error/success status, total and bounded pages
6	Code/name/hours, field errors, Save/Cancel
7	Saved/new timing rows, unit labels, Add, local Remove and staged Retire
8	Configured/current unit plus revision-sensitive stale warning and Confirm
9	Retirement/discard dialog, safe initial focus and return focus

5. Screen item definition and validation

Quoted labels are approved-BD proposals, not implemented catalog entries.

Item / Japanese UI (English gloss)	Local type / required	Validation and state rule	Source
q, search	string / no	Code/name search; input separate from applied URL; exact API bound in DD-API	BD 0-3, region 3
state / page	enum / integer	Active/retired/all; default active, page 1; Apply/Clear resets page; exact paging bound in DD-API	BD regions 3/5
id / parent version	opaque strings / edit only	UUID route id; authoritative parent xmin token; never increment/merge locally	DB 3.1/9
「ラインコード」 (Line code)	string / yes	Trim both ends; 1–50 Unicode code points; immutable on edit; duplicate comparison follows DB lower(code), all states	BD items; DB 3.1/4
「ライン名」 (Line name)	string / yes	Trim; 1–200 code points; not a URL/log label	BD items; DB 3.1
「稼働時間／日」 (Working hours/day)	decimal input text / yes	0 < value <=24, at most 3 fractional digits; show 「時間」 (hours)	BD items; DB 3.1
「生産可能な製品」 (Supported products)	product IDs / zero or more	New rows use active products; exclude every already associated ID including retired pairs; saved identity immutable	BD items; DB 3.2
「製造時間／単位」 (Production time/unit)	decimal input text / yes per row	0 < value <=999999999.999; at most 3 fractional digits; minutes per exactly one displayed unit	BD items; DB 3.2
Pair active state	boolean / saved row	Retirement one-way; saved retired rows readable, not re-added/reactivated; unsaved rows removable locally	BD retirement; DB 3.2

Item / Japanese UI (English gloss)	Local type / required	Validation and state rule	Source
Configured/current unit and revision	unit + opaque revision string	Seven units; compare unit and revision; changed-back unit remains stale; never use JS Number for bigint revision	DB 3.2/3.3/9
「現在の単位で確認」 (Confirm for current unit)	explicit local intent / conditional	Requires valid coefficient and displayed current unit/revision; no load/save auto-confirmation; server checks observed revision	BD unit confirmation; DB 7/9
「保存」 / 「キャンセル」 (Save / Cancel)	actions	Save logical aggregate; prevent repeated pending writes; Cancel writes nothing and guards dirty data	BD events

Keep decimal input as text; accept ASCII digits with at most one decimal point and 1–3 fractional digits if present. Reject exponent, thousands separators, NaN/infinity, nonpositive values and extra precision; do not round to fit. Parse and format with exact decimal helpers/server decimal, not binary floating-point arithmetic. Values display with Japanese formatting and visible unit labels; formatted thousands separators are not sent back as input. API transport details and authoritative parsing are finalized in DD-API/FN.

Dirty state includes pair additions, coefficient edits, retirement intent and confirmation intent. A saved stale pair may remain unchanged during an unrelated line edit; editing its timing requires explicit current-unit confirmation. New pairs require confirmation of the visible unit when saved. Retired pairs remain read-only. An inactive product's existing pair is readable but never eligible for new order selection/start. Changing product unit does not alter coefficients. A later coefficient or observed-unit change clears local confirmation intent. Save is disabled while clean, pending or unresolved unknown/conflict; preserve the draft until explicit verification/reload resolves those states. Confirmation covers the observed unit revision: a later product change must fail that intent rather than silently bind it to an unseen newer unit.

6. Visible states, permissions and dialogs

State / trigger	Visible behavior / retained data
List loading	Announce loading; no old rows presented as new filter results
No lines / no matches	Distinct notices; permitted New action remains available
Ready / retired	Readable actual rows, state text; retired line cannot be restored; code always immutable
Form loading / not found	No editable fake baseline; retry transient read, local 404 view
Clean / dirty form	Edit loaded data; dirty guard covers Cancel, navbar, browser navigation/reload
Validation	Preserve all values/rows; summary and associated errors; focus first invalid field
Stale unit	Show configured/current unit and required confirmation even when unit strings match but revision differs

State / trigger	Visible behavior / retained data
Conflict	Keep draft; explain stale record/unit; explicit guarded reload; no silent overwrite/merge
Saving / retiring	Disable repeated mutation; announce pending; do not erase draft before confirmed response
Success	Form returns to list; line retirement stays on list; announced notice and authoritative refresh
Unknown write outcome	Preserve draft and target identity; no automatic replay; read/list verification before another write
403 / 401	Forbidden stays on route without write; 401 uses existing session-expiry/login handling

Authenticated reads retain the applicable existing authorization policy; exact read-policy mapping is catalogued at DD-API review. No new read entitlement is introduced. Admin and Operator may create/edit/retire; API repeats these gates. A user without write permission has no write controls and a direct form route is forbidden, even if a read policy permits viewing. No broader role matrix is inferred. Retired-line ordinary field edits do not reactivate it; active state cannot be toggled back by form/API payload.

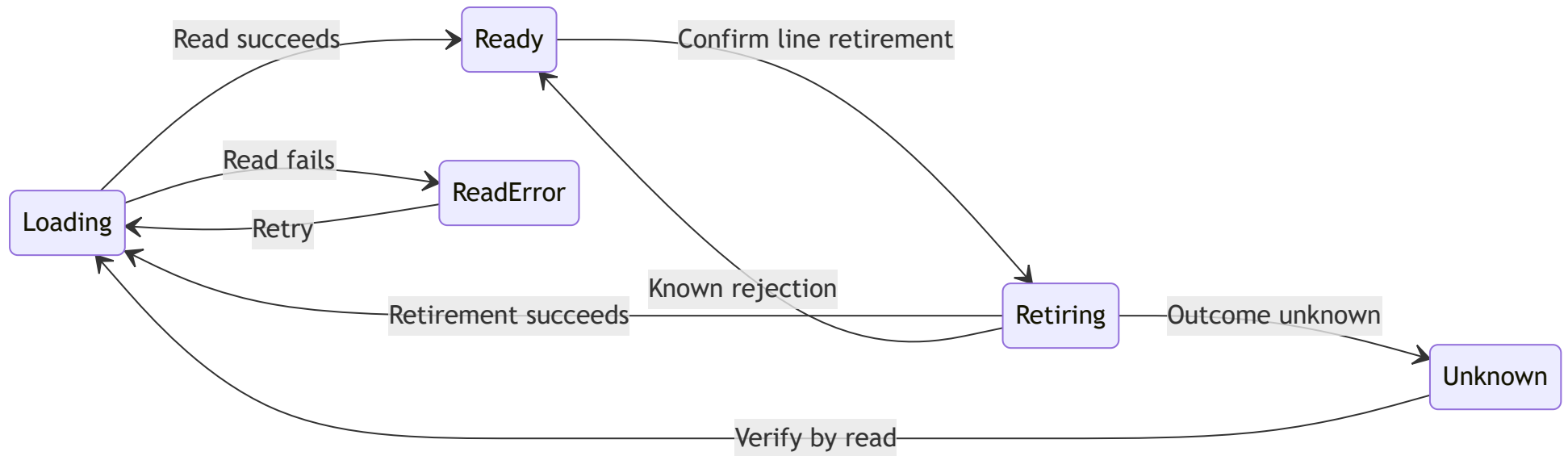
Pair retirement confirmation stages a local inactive intent; Save persists it with the entire aggregate. Cancel/Escape changes nothing; dirty confirmation does not commit other edits implicitly. Show pending retirement until Save succeeds. Line retirement is a separate list operation using the current parent version.

Dialogs center in both viewport axes with at least 16px margins, including after scroll. Safe action starts focused, background inert, Escape cancels, focus returns to invoker (or updated row/status after success). Fit SP and 200% zoom with internal scrolling. Labels/errors, semantic table/cards, live statuses, keyboard focus and visible non-color state cues target WCAG 2.2 AA.

7. Processing and state transitions

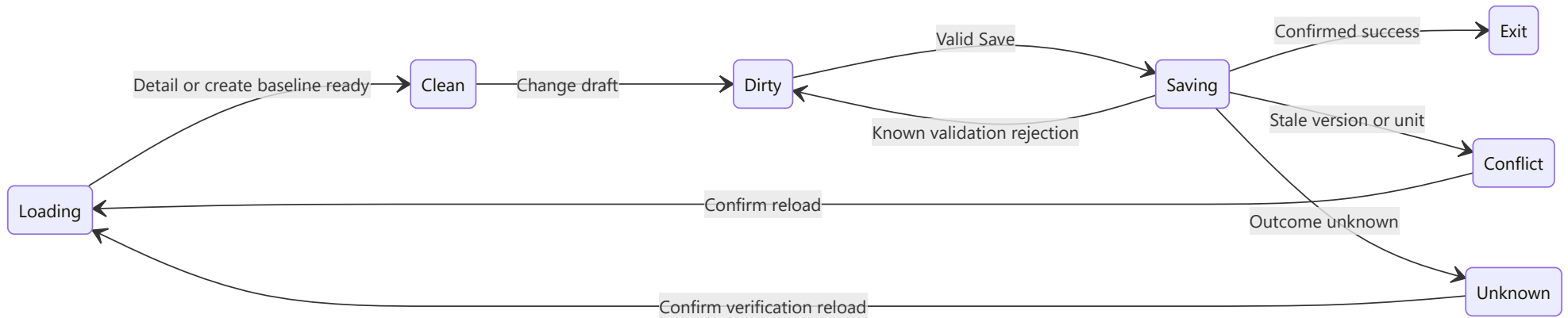
Distinct states are diagrammed below; same-state and any-state transitions are table-only. Retirement/discard dialogs are overlays with remembered return state, not separate routes. Entry/exit routes remain exactly those in the approved BD.

7.1 List state



From	Event	To	Side effect
Entry / Ready	Enter, Apply, Clear or page change	Loading	Read applied URL; no write
Loading	Successful read (including empty page)	Ready	Render current rows/empty state
Loading	Failed read	ReadError	Show retry; retain filter input
ReadError	Retry	Loading	New read only
Ready	Confirm line retirement	Retiring	One version-checked retirement request
Retiring	Confirmed success	Loading	Notice, authoritative list refresh
Retiring	Known rejection	Ready	Error; record not assumed changed
Retiring	Lost/ambiguous result	Unknown	No automatic write replay
Unknown	Verify	Loading	Read target/list; reconcile outcome before another mutation
Ready	New / Edit	Form Loading	Route change; no write

7.2 Form state



From	Event	To	Side effect
Loading	Detail succeeds / create initialized	Clean	Set authoritative baseline and version; no write
Clean	Edit/add/confirm/stage pair retirement	Dirty	Local draft only
Dirty	More edits / invalid local Save	Dirty	Keep values; validation errors, no HTTP mutation
Dirty	Valid Save	Saving	One logical aggregate mutation
Saving	Confirmed success	Exit	Return list and announce success
Saving	Known validation/business rejection	Dirty	Retain draft and field errors
Saving	Stale parent/product-unit observation	Conflict	No overwrite; draft retained
Saving	Lost/ambiguous result	Unknown	Do not automatically replay
Conflict	Explicit guarded reload	Loading	Discard only after confirmation, fetch latest
Unknown	Explicit guarded verification reload	Loading	Read existing id, or list/code after create; no implicit second create
Clean / Dirty	Cancel or permitted navigation	Exit	Dirty discard confirmation first; no write
Any form state	Read failure / 404	Same route error view	No fabricated data; retry where valid
Any authenticated state	401 / 403	Login / forbidden view	Existing session handling; no further mutation

The list chart has eight distinct-state transitions and the form chart nine; the rows enumerate them plus entry, navigation and same/any-state cases. Pending network response is not proof of commit. If access changes mid-edit, preserve values only in memory as allowed by existing session handling; no localStorage or URL draft persistence.

Processing flow owners

Flow	Later owner	Events / function
URL reads, request cancellation and focus	005_DD-SPD list block	Search/Clear/page, FN-032
Form load/dirty/validation/save	005_DD-SPD form block; 005_DD-FN aggregate write	FN-033
Product lookup, timing and confirmation	005_DD-SPD timing block; 005_DD-FN unit guard	FN-034
Dialog/list retirement	005_DD-SPD dialog block; 005_DD-FN retirement	FN-035
Order picker/start/history	New 005_DD-ORD; 005_DD-FN eligibility	FN-036

8. APIs used and persistence boundaries

Proposed call inventory below is for later DD-API review, not a finalized field catalog. Existing Product master endpoint/contract is preserved; product unit observation needed by this feature is resolved in the new feature's read contract. No separate pair DELETE, restore or independent-version mutation endpoint.

Proposed call	Purpose	Specification owner
GET /api/production-lines	Search/state/page list	005_DD-API
GET /api/production-lines/{id}	Line/pair detail and current unit observation	005_DD-API
POST /api/production-lines	Create line with configured pairs	005_DD-API
PUT /api/production-lines/{id}	Atomic parent/pair edit and explicit confirmations	005_DD-API
POST /api/production-lines/{id}/retire	Version-checked one-way line retirement	005_DD-API
Product choices / eligible order lines	Active choices, unit observation and eligibility projections	005_DD-API and new 005_DD-ORD; exact read route reserved

Operation	Persistence / transaction	Concurrency and history
List/detail	Project used fields from lines/pairs/products	Read-only; do not claim eligibility remains valid after response
Create/edit with pair changes	READ COMMITTED aggregate write	Product locks sorted UUID, then exclusive parent, then pairs; parent xmin changes on every persisted pair mutation

Operation	Persistence / transaction	Concurrency and history
Explicit confirmation	Same aggregate transaction; locked product observation	Compare observed unit/revision; server derives confirmed values; ABA invalidation retained
Retire line	Parent version comparison and exclusive lock	Preserve pairs/orders; no physical deletion
New/changed order assignment/start	Product, line, pair FOR SHARE through commit	Validate active/current-unit state and order concurrency; FK alone insufficient
Historical unchanged order edit	Preserve exact pair/null and existing field locks	No active check except Draft start; line locked outside Draft

Schema, columns, grants, indexes, trigger and recovery limits remain those of approved DB. Queries project only needed columns; read projections may join pairs/products/lines to report meaningful current units and historical names. This is a deliberate exception to separate-table query preference; avoid N+1 queries. Exact snapshot/read semantics and bounded query controls belong to API/FN. No scheduling, overload gate, automatic allocation or timing snapshot.

9. Exceptions and observability boundary

RFC 9457 error catalogs and exact time/lock limits are finalized in DD-API/FN; no unconfigured timeout is reported as working. Reads may be retried explicitly; writes are not automatically replayed after an ambiguous result.

Failure	Screen response / retry	Required observability
Validation / duplicate code	Retain fields, associate errors; user corrects before retry	Bounded validation/conflict outcome, no raw names/code/search
Stale parent / unit observation	Conflict with guarded reload; no overwrite	Operation/outcome and concurrency span
Not found / forbidden / unauthenticated	Local 404/403 or login per existing session	Status/outcome, no credentials or cookie content
Lock timeout / deadlock	Rollback result, preserve draft; explicit retry only after known failure	Bounded transient outcome and duration
Network / unknown commit	Keep draft; verify target/list before next mutation	Client operation/status without bodies; server outcome if known

Every new endpoint/use case requires OpenTelemetry spans and metrics, not logs alone. API/FN owns exact names and bounded labels; never include product/line IDs, raw text, coefficients, tokens or credentials as metric dimensions. Structured logs must not expose SQL, stack traces or sensitive payloads to users. Full security and instrumentation consistency gates remain pending companions.

10. Verification viewpoints and next review

Requirement	Later test viewpoint	Current verification state
REQ-064	URL restore, paging, empty/read-error and PC/SP list	Design mapping only; runtime not run
REQ-065	Trim/case duplicate, immutable code, parent stale write	Design mapping only; runtime not run
REQ-066	Exact bounds/precision; mixed-unit display; ABA/reconfirmation race; atomic pair edit	Design mapping only; runtime not run
REQ-067	Dialog Cancel no write; staged retirement/dirty guard; retained history; no restore	Design mapping only; runtime not run
REQ-068	Optional Draft, eligible start, outside-Draft lock and legacy null	Detailed order-impact design pending; runtime not run
REQ-069	Admin/Operator API/UI gates, keyboard/focus/dialog geometry, mobile/zoom and telemetry	Design mapping only; runtime not run

Test-plan IDs will be assigned in plan step 11; none are fabricated here. No business question is open. Exact API catalogs, query/time limits, backend flows, client sequences, detailed visual mockups and order impact remain at their planned review steps. Main-DD field/state consistency with approved BD/DB is reviewable; the full DD family and implementation-ready gate are incomplete. Submit this file and its EN/JA PDFs; wait for explicit review before DD-API. No approved design source was edited, application test run or migration executed.